

Homology: What Closes, and the Duality with Cohomology

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Abstract

Cohomology asks what persists under transformation. Homology asks what closes on itself. They are dual: the Universal Coefficients Theorem is the bridge, Poincaré duality is the isomorphism (on a closed orientable manifold), and the Kronecker pairing $H^n(X) \times H_n(X) \rightarrow \mathbb{Z}$ is the inner product between the two sides. This paper builds the homological dual of the cohomological body assembled in [2, 3, 4]: singular and simplicial homology, the Hurewicz theorem, Borel–Moore homology (the homology of non-compact spaces that pairs with compactly supported cohomology), motivic homology $H_n^{\mathcal{M}}(X, \mathbb{Z}(r))$ (the homological side of the motive $h_*(X)$), and the cycle class map as a homological statement. The functional zero point — where homology and cohomology meet in perfect duality — is Poincaré duality: the isomorphism $H^k(X) \cong H_{n-k}(X)$ for a closed n -manifold, the fulcrum that balances the two sides.

1 Homology: What Closes

Definition 1 (Singular homology). For a topological space X , the *singular chain complex* $C_*(X, \mathbb{Z})$ has $C_n(X) = \mathbb{Z}[\text{Hom}(\Delta^n, X)]$, the free abelian group on continuous maps from the standard n -simplex Δ^n to X . The boundary map $\partial_n : C_n \rightarrow C_{n-1}$ sends a simplex σ to the alternating sum of its faces: $\partial_n \sigma = \sum_{i=0}^n (-1)^i \sigma|_{[v_0, \dots, \hat{v}_i, \dots, v_n]}$. The *singular homology* is $H_n(X, \mathbb{Z}) = \ker(\partial_n) / \text{im}(\partial_{n+1})$: cycles (chains with no boundary) modulo boundaries (chains that are themselves the boundary of something).

Remark 2 (Cycles and coboundaries as dual operations). A cycle is a chain that closes: $\partial \sigma = 0$. A cocycle is a cochain that closes: $d\omega = 0$. A boundary is a cycle that is the edge of something: $\sigma = \partial \tau$. A coboundary is a cocycle that is the derivative of something: $\omega = d\eta$. Homology $H_n = \ker \partial / \text{im} \partial$: cycles modulo boundaries. Cohomology $H^n = \ker d / \text{im} d$: cocycles modulo coboundaries. The boundary map ∂ and the coboundary d are adjoint: $\langle d\omega, \sigma \rangle = \langle \omega, \partial \sigma \rangle$. This adjunction is Stokes’ theorem.

2 The Universal Coefficients Theorem

Theorem 3 (Universal Coefficients). *For a topological space X and abelian group A , there are natural short exact sequences:*

$$\begin{aligned} 0 \rightarrow H_n(X) \otimes A \rightarrow H_n(X, A) \rightarrow \operatorname{Tor}_1(H_{n-1}(X), A) \rightarrow 0, \\ 0 \rightarrow \operatorname{Ext}^1(H_{n-1}(X), A) \rightarrow H^n(X, A) \rightarrow \operatorname{Hom}(H_n(X), A) \rightarrow 0, \end{aligned}$$

both splitting (unnaturally). In particular: $H^n(X, \mathbb{Z}) \cong \operatorname{Hom}(H_n(X), \mathbb{Z}) \oplus \operatorname{Ext}^1(H_{n-1}(X), \mathbb{Z})$.

Remark 4 (The bridge). Universal Coefficients is the bridge between the two sides. The free part of H_n maps to H^n via Hom . The torsion part of H_{n-1} maps to H^n via Ext^1 (a shift: the torsion of H_{n-1} appears in H^n , not H^{n-1}). The Kronecker pairing $\langle \cdot, \cdot \rangle : H^n \times H_n \rightarrow \mathbb{Z}$ is the evaluation of a cocycle on a cycle: non-degenerate on the free parts.

3 Poincaré Duality: The Fulcrum

Theorem 5 (Poincaré duality). *Let X be a closed oriented n -manifold. Then for all k :*

$$H^k(X, \mathbb{Z}) \cong H_{n-k}(X, \mathbb{Z}).$$

The isomorphism is given by cap product with the fundamental class $[X] \in H_n(X, \mathbb{Z})$: $\alpha \mapsto \alpha \cap [X]$.

Remark 6 (Poincaré duality as the functional zero point). Poincaré duality is the point at which cohomology and homology are the same thing. Not approximately, not asymptotically — isomorphically. The lever of cohomology and the lever of homology meet at the fundamental class $[X]$. The fundamental class is the fulcrum: the point from which both sides are measured. It is not a forbidden zero. It is not missing. It is the zero that holds both sides in balance: the functional zero point of the duality.

4 Borel–Moore Homology

Definition 7 (Borel–Moore homology). For a locally compact space X with one-point compactification $X^+ = X \cup \{\infty\}$, the *Borel–Moore homology* is:

$$H_n^{BM}(X) = H_n(X^+, \{\infty\}).$$

For a closed oriented n -manifold: $H_n^{BM}(X) = H_n(X)$. For $X = \mathbb{R}^n$: $H_n^{BM}(\mathbb{R}^n) = \mathbb{Z}$ (one fundamental class), while $H_n(\mathbb{R}^n) = 0$ (no compact cycles). Borel–Moore homology pairs with compactly supported cohomology H_c^* via: $H_c^k(X) \times H_k^{BM}(X) \rightarrow \mathbb{Z}$.

Remark 8 (Why Borel–Moore matters). In algebraic geometry, algebraic cycles define classes in Borel–Moore homology, not in ordinary homology: an algebraic subvariety $Z \subset X$ of dimension r defines a class $[Z] \in H_{2r}^{BM}(X)$. The cycle class map of the Tate and Hodge conjectures [5] is the Borel–Moore cycle map: $\operatorname{cl} : \operatorname{CH}^r(X) \rightarrow H_{2d-2r}^{BM}(X)$ (where $d = \dim X$), identified with H^{2r} via Poincaré duality.

5 Motivic Homology

Definition 9 (Motivic homology). The *motivic homology* of a smooth variety X over k is

$$H_n^{\mathcal{M}}(X, \mathbb{Z}(r)) = \mathrm{CH}^{\dim X - r}(X, n - 2r),$$

the higher Chow group (Bloch [1]). The motivic cohomology $H_{\mathcal{M}}^n(X, \mathbb{Z}(r))$ and homology $H_n^{\mathcal{M}}(X, \mathbb{Z}(r))$ are related by motivic Poincaré duality for smooth projective X ($d = \dim X$): $H_{\mathcal{M}}^n(X, \mathbb{Z}(r)) \cong H_{2d-n}^{\mathcal{M}}(X, \mathbb{Z}(d-r))$.

Proposition 10 (Homological Beilinson). *The Beilinson conjectures [2] can be stated homologically: the motivic homology group $H_1^{\mathcal{M}}(X, \mathbb{Z}(1)) = \mathcal{O}(X)^\times$ (units), and the regulator map $r : H_n^{\mathcal{M}}(X, \mathbb{Z}(r)) \rightarrow H_D^{2r-n}(X)$ (Deligne cohomology) is the homological avatar of Beilinson's regulator. The leading coefficient of $L(X, s)$ at a critical value is (up to periods) the determinant of this regulator map.*

6 The Full Duality

Theorem 11 (Homology-cohomology as the complete picture). *The body of arithmetic geometry has two sides:*

<i>Cohomological</i>	<i>Homological</i>	<i>Bridge</i>
$H^n(X)$: what persists	$H_n(X)$: what closes	Universal Coefficients
Cocycles $d\omega = 0$	Cycles $\partial\sigma = 0$	Stokes' theorem
Cohomology class $[\omega]$	Homology class $[\sigma]$	Kronecker pairing
Motivic cohomology	Higher Chow groups	Motivic duality
Selmer group H_f^1	Mordell–Weil $E(\mathbb{Q})$	Kummer map
L -function zero: depth	Cycle class: dimension	Beilinson regulator
Poincaré duality: $H^k \cong H_{n-k}$		Fundamental class $[X]$

The Selmer group $H_f^1(G_{\mathbb{Q}}, V_p(E))$ is cohomological: it measures deformations of the Galois representation. The Mordell–Weil group $E(\mathbb{Q})$ is homological: it is the group of cycles (rational points, hence algebraic cycles of dimension 0) on E . The Kummer map $E(\mathbb{Q}) \otimes \mathbb{Q}_p \hookrightarrow H_f^1(G_{\mathbb{Q}}, V_p(E))$ is the bridge: it sends a cycle (a point) to a cocycle (a Galois cohomology class). BSD is the statement that the bridge is an isomorphism: the homological and cohomological sides have equal rank.

Remark 12 (The fulcrum of the body). Throughout this body of work, cohomology has been the primary language. The Selmer group, the étale cohomology, the motivic cohomology, the de Rham cohomology — all cohomological. Homology has been implicit: algebraic cycles, rational points, the Mordell–Weil group. The Kummer map makes the duality explicit. Poincaré duality on the Shimura variety [6] is the functional zero point where both sides meet. BSD is the statement that, at the fulcrum, the two sides balance.

References

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